

ORIGINAL ARTICLE

Optimizing student engagement detection using facial and behavioral features

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Abstract

In recent years, computer vision and machine learning have achieved remarkable advancements across health care, autonomous systems, and robotics sectors. However, the educational domain, particularly the automated detection of student engagement in online and offline learning environments, remains rich in research opportunities. Detecting student engagement is inherently complex and requires sophisticated interpretive capabilities. This paper introduces a novel approach to automatically detect and classify student engagement by integrating facial images with behavioral and facial features, providing a comprehensive solution to enhance engagement recognition. This study begins by extracting behavioral features and correlating them with pre-defined engagement labels to perform engagement classification using machine learning techniques. In parallel, deep learning models are trained and validated on both image and behavioral features, offering a complementary approach. Additionally, the relationship between facial action units (AUs) and engagement labels is analyzed using three distinct metrics: conditional activation probability, relative activation ratio, and the statistical discriminant coefficient (SDC). The study utilizes publicly available datasets (WACV and DAiSEE) to perform extensive evaluations. Experimental results demonstrate that integrating action units significantly enhances model performance, with XGBoost achieving the highest accuracy (82.9%) among traditional models and EfficientNet reaching the best performance (47.2%) in deep learning experiments. These findings highlight the potential of multimodal approaches in improving real-time engagement detection, offering valuable insights into educational technologies and pedagogy.

Keywords Student engagement detection · Facial expression · Facial action unit · Behavioral engagement · Affective computing

1 Introduction

In a teaching-learning environment, student participation plays a crucial role as a pivotal factor, exerting a profound influence on diverse facets of learning outcomes and classroom dynamics. Fredricks et al. [1] underline its significance in learning maintenance, academic achievement, dropout rates, boredom, frustration, alienation, as well as student stress and depression. However, assessing individual student engagement in larger classes, especially in online teaching platforms, poses challenges for instructors. The transition to online and hybrid teaching modes post-COVID-19 has prompted the exploration of automated methods to detect and recognize student engagement, using recorded image or video data from virtual platforms or classroom cameras.

However, evaluating individual student engagement within larger classes, particularly in online teaching platforms, presents a daunting task to instructors. Automated student engagement detection offers a solution to



this challenge by alleviating the burden of manual monitoring, thereby allowing educators to dedicate more time to instructional and learning activities enabling educators to dedicate more time to teaching and learning activities. Automated detection of student engagement offers significant benefits for educational stakeholders, including instructors, students, parents, and decision-makers. For instructors, it provides insights into areas where students may require additional support, facilitating targeted interventions to enhance engagement and improve instructional quality. Furthermore, it streamlines the oversight process, permitting educators to concentrate more fully on pedagogical tasks. Students benefit from improved learning outcomes, while parents can support their children's education more effectively. Decision-makers can use engagement data to inform course design and institutional strategies, ultimately enhancing student satisfaction and institutional reputation.

This study addresses the recognition of student participation through facial expressions, an area of ongoing research that employs machine learning and deep learning techniques [2, 3]. While previous research has mapped basic emotions to facial features, there is a notable gap in mapping facial features directly to engagement labels. To bridge this gap, this paper performs a comprehensive analysis of both machine learning and deep learning approaches to establish a connection between facial action units (AUs) and engagement labels. By utilizing advanced tools such as OpenFace [4] and MediaPipe [5], and leveraging the publicly available WACV dataset, various algorithms are explored for this purpose.

The main contributions of this study include:

- Mapping facial action units (AUs) to engagement labels.
- Conducting a systematic analysis of both traditional machine learning and deep learning approaches to estimate engagement from images and videos.
- Providing a reproducible code repository for this work.¹

The remainder of the paper is organized as follows. Section 2 provides an overview of recent developments in student engagement detection, beginning with traditional machine learning approaches that utilize basic facial expressions and extending to deep learning techniques that offer more advanced and accurate models for engagement recognition. Section 3 details the dataset and the facial feature extraction methods employed in this study, including descriptions of the student engagement dataset, the extraction of facial features using OpenFace and MediaPipe, and an explanation of key features such as action units, eye gaze, head pose estimation, and facial landmarks generated by MediaPipe Face Mesh. Section 4 elaborates on the experimental setup, particularly addressing the handling of imbalanced datasets, and provides insights into the mapping of action units to engagement labels. It also compares the effectiveness of machine learning and deep learning approaches in engagement classification, covering methods such as input concatenation and the dual-path feature approach. Section 5 presents a discussion on the interpretability of the models, including a detailed analysis of the results, an ablation study, and a review of hyperparameter tuning. Section 6 provides a comparative analysis of action unit mapping in static images and dynamic videos, highlighting methodological differences and implications for improving engagement detection performance in dynamic educational environments. Finally, Sect. 7 concludes the paper by summarizing the findings, discussing their practical implications, and suggesting directions for future research.

2 Related work on student engagement recognition

Since early 1980, engagement detection has been treated as a major research problem. Engagement detection was first introduced by [6], where Hernandez et al. performed engagement analysis on a small private customized dataset of TV viewers. In [7], engagement detection has been categorized into three categories: automatic, semiautomatic, and manual. The manual category is either self-reporting via questionnaires or observation by an

¹ In the spirit of reproducible research, the code is available here: <https://github.com/riju-jec/Student-engagement>.

expert to note down the learner's engagement labels. In the semiautomatic category, learners' indirect involvement and probabilistic inference are used to determine the engagement labels. And the automatic category has become the most popular in recent years. Many researchers have been working on automatic engagement detection based on different modalities like facial expression, eye tracking, e-learning tools log files, body postures, sensory data (EEG, heart rate, blood pressure, etc.) [8], etc.

Student engagement is a multifaceted concept, which consists of behavioral, emotional, and cognitive engagement. These three domains are separate yet interrelated. In this study, the primary concentration is placed on behavioral and emotional engagement. Facial expression plays a significantly vital role in the recognition of behavioral and emotional engagement. By observing someone's facial expression, the human brain can automatically detect whether the person is feeling happy, sad, disgusted, angry, or paying attention or not. But for a computer or machine to automatically detect and recognize these emotions, analyzing facial expressions is challenging. With the advancement of artificial intelligence and computer vision, researchers have been working on designing various machine-learning models to understand human emotions. One of the simplest ways of recognizing facial emotion is by studying facial images or video clips. Whitehill et al. [9] performed binary engagement classification using GentleBoost with Box Filter features (BoostBF) [10], support vector machines with Gabor features (SVM-Gabor) [11], and multinomial logistic regression with expression outputs from the Computer Expression Recognition Toolbox (CERT) [11]. They concluded that SVM-Gabor outperforms the other two methods with better accuracy. Bosch et al. [12] employed CERT to obtain FACS facial features, which were then used for building classification models to detect the engagement levels: confusion, frustration, boredom, neutral, and engaged. The author performed further research [13, 14] on engagement detection using other methods like FACET [13], a commercial version of CERT, to obtain a larger set of action units (AU) to work on.

In recent years, deep learning approaches have emerged as powerful tools for engagement recognition [15]. Abedi and Khan [16] proposed a novel method for automatic engagement measurement in virtual learning environments, using affective states and ordinal classification to achieve significant accuracy improvements. Savchenko et al. [17] introduced a pipeline utilizing a single efficient neural network for real-time classification of student emotions and engagement. Gupta et al. [18] developed a deep learning-based system for real-time engagement detection in online learning through facial emotion recognition. They evaluated models such as Inception-V3, VGG19, and ResNet-50 on benchmark datasets like FER-2013, CK+, and RAF-DB. ResNet-50 achieved the highest accuracy in classifying engagement states as "Engaged" or "Disengaged". Zhang et al. [19] introduced a self-supervised learning approach for engagement recognition using a facial masked autoencoder (FMAE). This method handles large-scale unlabeled facial video data and demonstrated strong performance on DAiSEE and EmotiW datasets by focusing on key facial regions and capturing accurate spatiotemporal details. Sukumaran and Manoharan [20] proposed a multimodal engagement recognition system that integrates facial emotion recognition, gaze tracking, head pose estimation, and eye blink rate using deep learning techniques. The effectiveness of this system was validated through formative assessment quiz scores, showing its practical applicability in online learning environments. Copur et al. proposed an engagement detection with multitask training (ED-MTT) system [21] and Huang et al. [22] proposed the deep engagement recognition network (DERN) to recognize student engagement from facial features related to facial AUs, eye and head movement positions. Additionally, a comprehensive literature review conducted by Mandia et al. [15] highlighted a diverse array of machine learning methodologies—including facial expression analysis, eye movement tracking, speech pattern evaluation, keystroke dynamics monitoring, and deep learning approaches such as CNNs, LSTMs, and ensemble learning techniques—underscoring the complexity and broad scope of current methods utilized in automatic student engagement detection.

Beyond emotion detection, the challenge of mapping engagement directly to facial features, specifically facial action units (AUs), has been explored. Traditional approaches, such as those by Sharma et al. [23], calculated a concentration index (CI) based on basic emotions, which was then used to label student engagement. Bhardwaj et al. [24] proposed the mean engagement score (MES), which similarly relied on basic emotion probabilities and their corresponding weights. However, recognizing engagement, a subtler affective state, purely based on basic

emotions like happiness, sadness, or anger is not always feasible in real-time applications. Therefore, in this paper, rather than focusing solely on basic emotions, the engagement labels have been mapped directly to facial features such as facial AUs. Previous studies [25, 26] have proposed mapping between basic emotions and facial action units. More recently, Zhao et al. [27] introduced a lightweight model (LightFER) that improved real-time engagement detection by efficiently classifying facial expressions and mapping basic emotions directly to engagement labels. Despite these advancements, to the best of our knowledge, no existing work directly maps engagement labels to facial action units (AUs). To address this research gap, the present study proposes a method that bypasses the intermediate step of basic emotion classification by directly mapping student engagement labels to facial AUs, thereby enabling more robust, precise, and real-time engagement detection.

3 Dataset and facial features used for engagement recognition

3.1 Student engagement dataset

In this study, the **WACV** dataset, provided by Kamath et al. [28], was utilized. This dataset was selected for its notable attributes. Firstly, it offers a substantial sample size and is publicly accessible. Secondly, it captures student images and videos in authentic classroom settings with background noise, ensuring genuine facial expressions. Additionally, each image in the dataset is annotated with three engagement classes, facilitating classification tasks. The WACV dataset consists of 4424 student images of various dimensions, which were standardized to (100,100,3). These images are categorized into three classes: disengaged (class 0), partially engaged (class 1), and engaged (class 2). The dataset has an imbalanced class distribution, with 412, 2247, and 1765 images for disengaged, partially engaged, and fully engaged classes, respectively. Figure 1 shows sample images of these three engagement classes. All images used in this study were obtained from the publicly available WACV dataset [28], which explicitly permits usage for academic and research purposes.

For extended analysis, another publicly available dataset, **DAiSEE**, was incorporated to facilitate AU mappings on video data. Further insights into the extended methodology are expounded upon in Sect. 6.

3.2 Facial feature generation from OpenFace and MediaPipe

Each of the different facial features like facial landmark features, eye gaze, head pose, and facial action units plays a crucial role in establishing student engagement. Although different facial feature extraction methods have been found in the literature, very few have provided their implementations in open-source code form. In this study, two state-of-the-art open-source tools—OpenFace [4] and MediaPipe [5]—were selected for their simplicity and accessibility to extract facial features like facial landmarks, eye gaze, head pose, and action units.

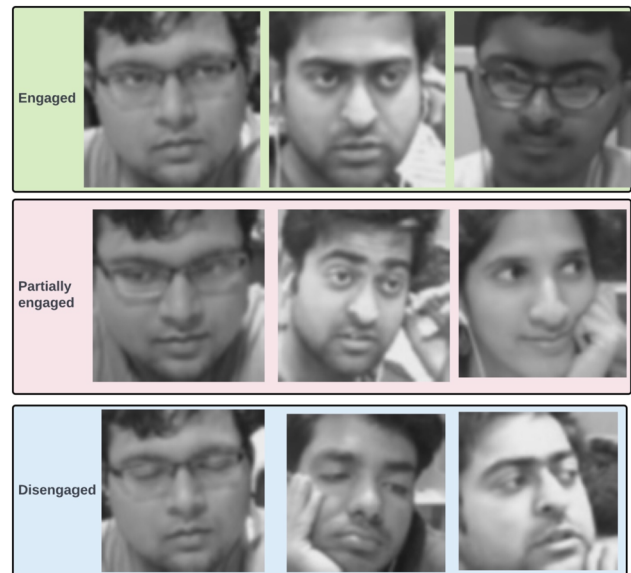
This section describes the different facial features extracted using OpenFace and MediaPipe that were utilized in this study.

3.2.1 Facial action units (AU)

OpenFace incorporates pre-trained models designed to recognize action units (AUs), which represent fundamental facial movements capable of describing a wide range of expressions. These AUs are defined by the Facial Action Coding System (FACS) [29], a widely accepted framework for characterizing facial expressions based on the underlying muscle movements. Each AU corresponds to specific facial muscle actions, and complex expressions often involve the activation of multiple AUs simultaneously.

In its output, OpenFace provides both the presence and intensity values of AU. The presence of AU indicates whether a particular AU is active (1) or inactive (0), while the intensity of AU is a continuous scale ranging from 0 to 5, where 0 signifies minimal intensity and 5 indicates maximum intensity. OpenFace is capable of

Fig. 1 Sample images of three different classes (disengaged, partially engaged, and engaged) from the WACV dataset



recognizing 17 AUs, including AU01-inner brow raiser, AU2-outer brow raiser, AU4-brow lowerer, AU5-upper lid raiser, AU6-cheek raiser, AU7-lid tightener, AU9-nose wrinkler, AU10-upper lip raiser, AU12-lip corner puller, AU14-dimpler, AU15-lip corner depressor, AU17-chin raiser, AU20-lip stretcher, AU23-lip tightener, AU25-lip part, AU26-jaw drop, and AU45-blink.

3.2.2 Eye gaze and head pose estimation

For facial landmark detection, OpenFace uses CLNF (constrained local neural fields) [30]. To improve the accuracy of facial landmark detection, CLNF uses a non-uniform regularized mean-shift method, which considers Local Neural Patch (LNP). For each image input, OpenFace provides 68 different 2D landmark locations (x and y coordinates), as well as 3D landmark locations (x, y, and z coordinates). OpenFace uses the facial landmark locations (3D and 2D) for the estimation of eye gaze and head pose. Key features utilized in this work include gaze direction vectors for both eyes in world coordinates ($gaze_{x0}$, $gaze_{y0}$, $gaze_{z0}$, $gaze_{x1}$, $gaze_{y1}$, $gaze_{z1}$), and head location and rotation relative to the camera ($pose_{Tx}$, $pose_{Ty}$, $pose_{Tz}$, $pose_{Rx}$, $pose_{Ry}$, $pose_{Rz}$).

3.2.3 Facial landmarks using MediaPipe face mesh

MediaPipe is an open-sourced tool developed by Google researchers [5]. The MediaPipe Face Mesh tool can be used to extract 468 3D facial landmark values from any image or video in real-time. For each image from the WACV dataset, the coordinates values (x, y) of 468 facial landmarks, $\{(x_i, y_i) | i = (0, 1, \dots, 467)\}$ have been extracted. Then these extracted facial landmark data of each image have been mapped with the corresponding engagement labels.

4 Experiments

4.1 Experimental setup

The experiments were conducted in two distinct settings: traditional machine learning (ML) and deep learning (DL). Both sets of experiments were executed on an NVIDIA A100 Tensor Core GPU node, equipped with

CUDA 12.2 and Driver 535.54.03. A Conda environment configured with PyTorch was established for the deep learning experiments to facilitate the computational requirements.

In addition, Optuna was employed for hyperparameter tuning to enhance the performance of the DL models. Optuna utilizes Bayesian optimization, which is more efficient than traditional hyperparameter tuning methods such as grid search and random search due to its ability to model the performance of hyperparameter configurations and prioritize promising regions of the search space. The tuning process involved defining a search space for critical hyperparameters, including learning rate, batch size, dropout rate, weight decay, etc. Optuna's Tree-structured Parzen Estimator (TPE) sampler was used to explore this search space.

Optuna was configured to run multiple trials in parallel, leveraging the computational capabilities of the A100 GPU. Each trial involved training the model with a different set of hyperparameters and evaluating performance using loss and validation accuracy. Early stopping was applied to terminate trials that did not show promise, thus conserving computational resources and focusing on the most promising configurations.

4.2 Dealing with the imbalanced dataset

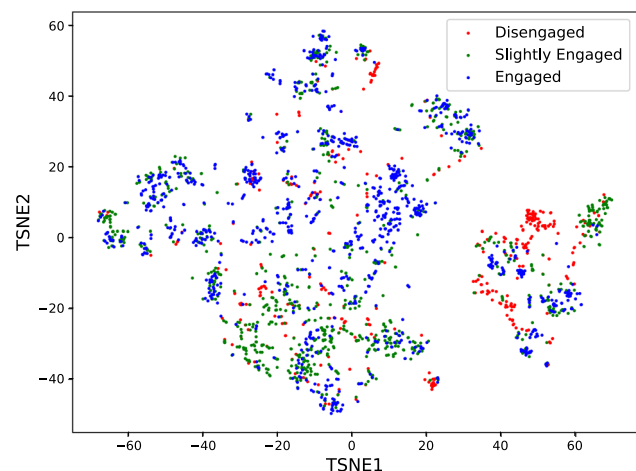
As mentioned in Sect. 3.1, the WACV dataset utilized in this study exhibited a significant class imbalance. Specifically, the dataset consisted of 412 disengaged images (class 0), 2247 partially engaged images (class 1), and 1765 engaged images (class 2). To visualize the imbalanced nature of WACV, t-SNE (t-distributed stochastic neighbor embedding) was employed to visualize the high-dimensional data in a two-dimensional space, as shown in Fig. 2. This imbalance posed a challenge as it could lead to a model biased toward the majority classes, thereby compromising the model's ability to accurately classify the minority class. To address this issue and ensure effective learning from all classes, a combination of techniques involving class weight computation and sampling strategies was implemented. These strategies were designed to mitigate the effects of class imbalance, thus enhancing the model's performance and reliability across all classes.

4.2.1 Class weight computation

The first step in handling the imbalanced dataset involved calculating the class weights. These weights were computed based on the frequency of each class in the complete set of training labels. The formula used for computing the weight of a class i is as follows:

$$Weight_i = \frac{N}{N_i} \quad (1)$$

Fig. 2 Distribution of the three engagement classes: disengaged (class 0, in red), partially engaged (class 1, in green), and engaged (class 2, in blue). The t-SNE algorithm reduces the high-dimensional data to two dimensions (TSNE1 and TSNE2), revealing the clustering of samples and highlighting the class imbalance present in the dataset



where N is the total number of samples and N_i is the number of samples in class i . This ensured that minority classes received higher weights, thereby emphasizing their importance during training.

Incorporation into Loss Function: To ensure the model paid appropriate attention to all classes, the computed class weights were incorporated into the loss function. Specifically, a weighted cross-entropy loss was utilized, which adjusted the loss contribution of each sample based on its class weight.

$$\text{Loss} = - \sum_{i=1}^C \text{Weight}_i \cdot y_i \cdot \log(\hat{y}_i) \quad (2)$$

where

- C is the total number of classes,
- $\text{Weight}_i = \frac{N}{N_i}$, N is the total number of samples, and N_i is the number of samples in class i ,
- y_i is the ground-truth binary indicator (0 or 1) indicating the correct classification for class i ,
- $\hat{y}_i$ is the predicted probability for class i .

This modification to the loss function helped in penalizing the misclassification of minority class samples more than that of majority class samples, thereby steering the model toward a more balanced performance.

4.2.2 *WeightedRandomSampler* for data loading

In addition to modifying the loss function, a *WeightedRandomSampler* is employed to balance the training data during the mini-batch selection process. The *WeightedRandomSampler* from PyTorch was utilized to address dataset imbalance during mini-batch creation. This approach calculates sampling weights inversely proportional to class frequencies, using the following method:

$$\text{Sample Weight}_i = \frac{1}{N_i} \quad (3)$$

where N_i represents the number of samples belonging to class i .

Each training instance from class i is assigned this weight, guiding the sampler to select minority class samples more frequently. This ensures each mini-batch contains a representative sample distribution from all classes, reducing potential bias toward majority classes during training.

4.3 Statistical analysis for AU engagement mapping

To establish a comprehensive relationship between action units (AUs) and engagement labels, we conducted a detailed statistical analysis across both static image (WACV) and dynamic video (DAiSEE) datasets. Our analysis employed three key coefficients: conditional probability, relative activation ratio, and the log-ratio statistical discriminative coefficient (SDC). These metrics were used to assess the significance of each AU in differentiating engagement states.

4.3.1 Conditional probability

The conditional probability quantifies how frequently an AU is activated within a specific engagement class. For a given AU j and engagement label y_i , the conditional probability is defined as:

$$P_{ij} = P(\text{AU}_j | y_i) \quad (4)$$

where

- P_{ij} is the conditional probability of AU j being active given engagement class y_i ,
- $P(\text{AU}_j|y_i)$ represents the proportion of frames (or images) labeled as y_i in which AU j was active.

This measure provides insights into which AUs are most commonly associated with each engagement state.

4.3.2 Relative activation ratio (RAR)

The relative activation ratio measures how distinctively an AU is associated with a particular engagement label by comparing its activation probability within that class to its overall activation probability. The relative ratio for AU j in engagement class y_i is given by:

$$R_{ij} = \frac{P(\text{AU}_j|y_i)}{P(\text{AU}_j)} \quad (5)$$

where

- R_{ij} is the relative activation ratio for AU j in engagement class y_i ,
- $P(\text{AU}_j|y_i)$ is the conditional probability as defined earlier,
- $P(\text{AU}_j)$ is the overall probability that AU j is active, calculated across all engagement classes.

An $R_{ij} > 1$ suggests that AU j is more activated in engagement class y_i than in the dataset overall, indicating its potential significance for distinguishing that class. Conversely, $R_{ij} < 1$ suggests that the AU is less frequently activated in class y_i compared to its baseline.

4.3.3 Log-ratio statistical discriminative coefficient (SDC)

The log-ratio SDC extends the concept of feature importance by quantifying the overall discriminative power of an AU across all engagement classes. It evaluates how distinctively an AU deviates from its baseline activation rate and is computed as:

$$S_j = \sum_i P(y_i) \log \left(\frac{P(\text{AU}_j|y_i)}{P(\text{AU}_j)} \right) \quad (6)$$

where

- S_j is the log-ratio statistical discriminative coefficient for AU j ,
- $P(y_i)$ is the prior probability of engagement class y_i ,
- $P(\text{AU}_j|y_i)$ is the conditional probability of AU j being active in engagement class y_i ,
- $P(\text{AU}_j)$ is the overall activation probability of AU j .

The log-ratio SDC is particularly informative in identifying AUs that deviate significantly from their baseline activation in relation to specific engagement classes. Higher absolute values of S_j indicate that AU j exhibits either heightened or suppressed activation relative to its overall baseline, thereby reflecting stronger discriminative potential.

Each of the three coefficients employed in our analysis offers distinct insights into the relationship between action units (AUs) and engagement states. The conditional probability P_{ij} quantifies the absolute occurrence of AU activations within a specific engagement class, providing a direct measure of AU presence in different states. The relative activation ratio (RAR) R_{ij} extends this understanding by comparing an AU's activation rate within a particular engagement class to its overall baseline activation, thereby identifying AUs that disproportionately

activate in certain engagement states. Meanwhile, the log-ratio statistical discriminative coefficient (SDC) S_j consolidates these insights by weighting each AU's deviation from its baseline activation with respect to the class probabilities, ultimately identifying the AUs with the strongest discriminative potential across all engagement classes.

These coefficients were initially derived from the WACV dataset, which comprises static images, and later extended to the DAiSEE dataset, where video frames were treated as individual static samples. This methodological alignment ensures consistency in AU-based mapping across both static and dynamic data contexts. The insights gained from this analysis have been integrated into our engagement detection framework to prioritize AUs that exhibit robust discriminative characteristics, ultimately improving model performance.

To visualize these mappings, Figs. 3, 4, 5 collectively illustrate the significance of individual AUs in characterizing engagement states. Figure 3 depicts the conditional probability values, which demonstrate the frequency with which each AU is activated within a given engagement class. Higher values indicate AUs that are consistently present in specific engagement states. Figure 4 presents the relative activation ratio (RAR), which highlights AUs that activate disproportionately in specific engagement classes relative to their baseline occurrence. Notably, AUs with higher RAR values are strong indicators of particular engagement states. Finally, Fig. 5 visualizes the log-ratio SDC values, representing the overall discriminative power of each AU across all engagement classes. Higher SDC values identify AUs that distinctly characterize specific engagement states, reinforcing their relevance in distinguishing between disengaged, partially engaged, and engaged behaviors.

Based on these visualizations, the top three AUs identified for each engagement label are as follows:

- Disengaged: AU45, AU25, and AU20 - These AUs exhibit strong activation patterns in the disengaged class, with AU45 demonstrating particularly high RAR and SDC values, marking it as a key marker for disengagement.
- Partially engaged: AU02, AU15, and AU20 - AU02 and AU15 show increased activation ratios and notable SDC values in the partially engaged class, suggesting their relevance in intermediate engagement states.
- Engaged: AU14, AU23, and AU20 - AU14 and AU23 consistently show distinct conditional probabilities and moderate to high RAR values in the engaged class, indicating their strong presence in positive engagement states.

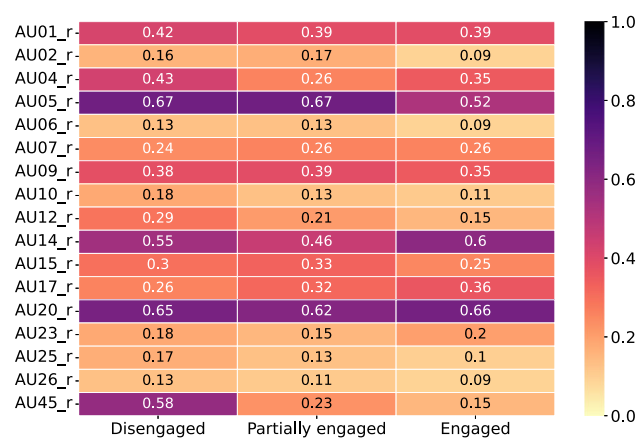


Fig. 3 Conditional probability mapping of AUs with engagement labels. This heatmap visualizes the probability of each AU being activated within each engagement class. For example, **AU05** and **AU20** maintain high probabilities across all engagement states, indicating their general presence in facial expressions. Conversely, AUs such as **AU45** show distinct patterns, with a higher activation probability in the disengaged class (0.58) compared to partially engaged (0.23) and engaged (0.15), highlighting its potential as a marker for disengagement

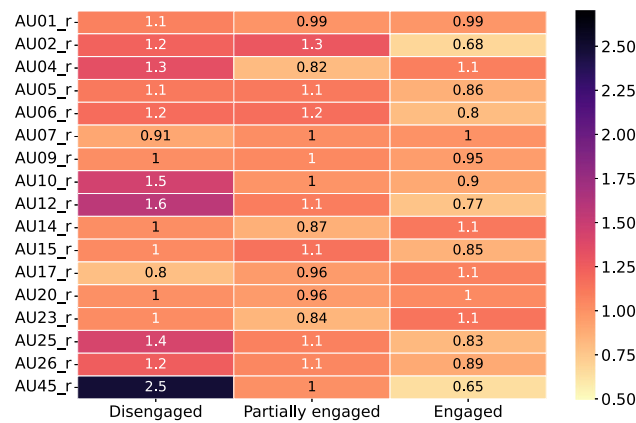


Fig. 4 Relative activation ratios for AUs across engagement labels. This heatmap displays the relative activation ratio, which identifies AUs that are disproportionately activated in specific engagement states. For instance, **AU45** exhibits a significant increase (2.5) in the disengaged class, reinforcing its association with disengagement. Conversely, **AU02** shows a stronger presence in the partially engaged class (1.3), suggesting its relevance for identifying partially engaged

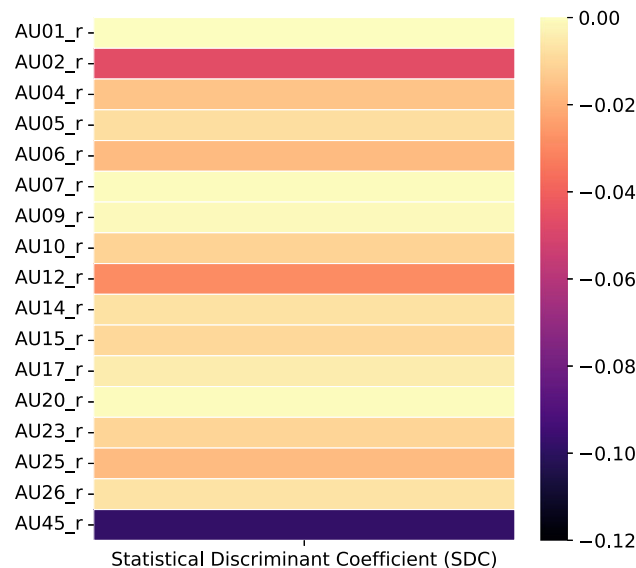


Fig. 5 Log-ratio statistical discriminative coefficient (SDC) Scores for AUs. This heatmap illustrates the overall discriminative power of each AU across all engagement classes. Higher absolute SDC values indicate that the AU is activated more prominently in specific engagement classes compared to its baseline. For instance, AU45 and AU02 exhibit notably higher SDC values, reinforcing their strong discriminative influence. Conversely, AUs such as AU12 and AU25 demonstrate moderate discriminative power, while AUs like AU01, AU07 and AU20 show minimal discriminative behavior across engagement states

This comprehensive mapping reveals that certain AUs, such as AU20, exhibit relevance across multiple engagement states, suggesting its involvement in broader facial dynamics, but its uniform activation yields low discriminative power. Conversely, AUs like AU45 show strong class-specific behavior, reinforcing their potential as key discriminators in student engagement detection.

4.4 Engagement classification: machine learning approach

The investigation of the WACV dataset involved meticulous evaluation of the performance of five prominent machine learning (ML) models-support vector machine (SVM), random forest, decision tree, gradient boosting, and XGBoost-for multiclassification tasks. A comprehensive suite of evaluation metrics was employed, including accuracy, precision, recall, F1-score, and analysis of the receiver operating characteristic (ROC) curve.

The analysis was structured by partitioning the dataset into four distinct feature sets: facial landmark features (S_1), eye gaze and head pose features (S_2), facial action unit (AU) features (S_3), and a composite feature set (S_4) comprising all available features. Figure 6 shows the graphical representation of the proposed pipeline. Each set of features underwent independent training of the ML models, enabling a comprehensive evaluation of their efficacy and suitability for the task at hand. To ensure unbiased evaluation, the entire datasets (S_1 , S_2 , S_3 , S_4) were partitioned into training (75%) and testing (25%) sets. Machine learning models were trained on the training datasets. Finally, once a trained model was obtained, it was evaluated on the unbiased test dataset.

The findings, as presented in Table 1, unequivocally demonstrate the superior performance of XGBoost on the test datasets of S_1 , S_2 , and S_3 , while gradient boosting excelled on the test dataset of S_4 . Furthermore, the visual comparison provided in Fig. 7 illustrates the area under the ROC curve for the five ML models: support vector machine (SVM), random forest, decision tree, gradient boosting, and XGBoost models across four distinct datasets S_1 , S_2 , S_3 , and S_4 . The ROC AUC values were computed to assess the classification performance of each model, with higher values indicating better discriminative capabilities. From the graph, it is evident that SVM exhibits the lowest performance across all datasets, while XGBoost and gradient boosting consistently outperform other models.

4.5 Engagement classification: deep learning approach

In addition to traditional machine learning models, deep learning approaches were explored for engagement classification using the WACV dataset. Specifically, several deep learning models were designed and evaluated, including variants based on ResNet and EfficientNet architectures. These models were implemented with two unique strategies: input concatenation and dual-path feature. The deep learning experiments involved two primary setups: models that utilize only image data (f_{img}) and models that incorporate both image data and OpenFace-derived features (f_{iof}).

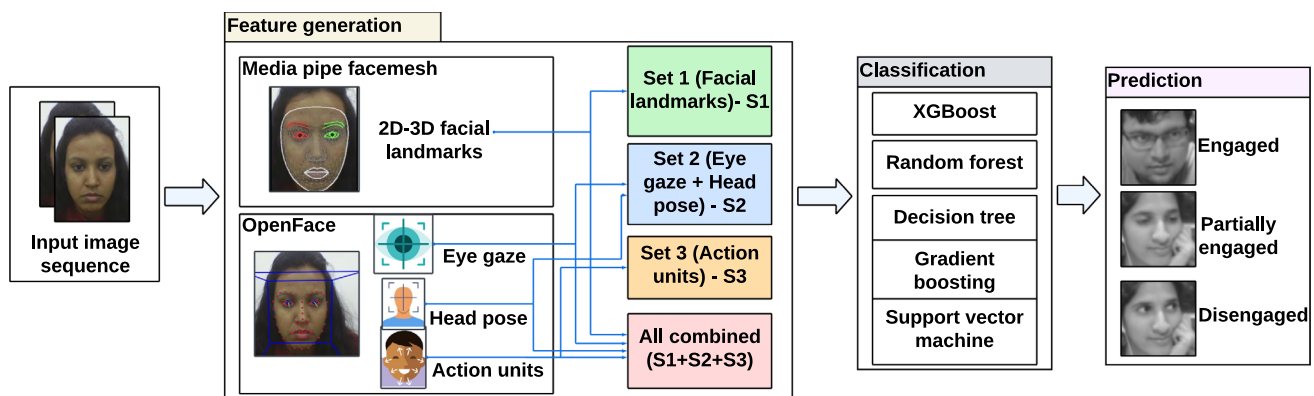


Fig. 6 The figure depicts the pipeline for multiclassification tasks on the WACV dataset. The pipeline involves the partitioning of the dataset into four distinct feature sets: facial landmark, eye gaze-head pose, action units, and all combined, followed by independent training of machine learning models-support vector machine (SVM), random forest, decision tree, gradient boosting, and XGBoost-on each feature set

Table 1 Evaluation results of different classification methods based on three different features: a. AU features, b. eye gaze and head pose features and facial landmarks points. The best evaluation metric values are highlighted with blue color

	Method	Accuracy	Precision	Recall	F1-score
Facial landmark features S_1	Support Vector Machine	0.345	0.409	0.339	0.345
	Random Forest	0.814	0.819	0.814	0.814
	Decision Tree	0.786	0.790	0.782	0.786
	Gradient Boosting	0.814	0.819	0.810	0.814
	XGBoost	0.823	0.829	0.819	0.821
Eye gaze and Head Pose features S_2	Support Vector Machine	0.353	0.118	0.333	0.353
	Random Forest	0.762	0.765	0.760	0.762
	Decision Tree	0.758	0.765	0.754	0.758
	Gradient Boosting	0.762	0.769	0.759	0.762
	XGBoost	0.802	0.803	0.799	0.802
AU features (Intensity) S_3	Support Vector Machine	0.353	0.118	0.333	0.353
	Random Forest	0.818	0.819	0.818	0.817
	Decision Tree	0.770	0.788	0.787	0.770
	Gradient Boosting	0.794	0.796	0.795	0.794
	XGBoost	0.829	0.830	0.830	0.829
Combined features S_4 ($S_1 + S_2 + S_3$)	Support Vector Machine	0.345	0.409	0.339	0.345
	Random Forest	0.825	0.837	0.825	0.825
	Decision Tree	0.806	0.810	0.805	0.806
	Gradient Boosting	0.833	0.835	0.834	0.833
	XGBoost	0.814	0.814	0.813	0.814

4.5.1 Input concatenation

As mentioned above, the models with data f_{img} are straightforward, employing the architecture of ResNet18 and EfficientNet-B0 pre-trained on the ImageNet dataset, with the final classifier layers replaced with engagement classes. The final fully connected layer was replaced to match the number of engagement classes (3). The rest of the network remained unchanged, allowing the model to focus on extracting relevant features from the images alone, as illustrated in Fig. 8. To improve generalization, dropout layers were added after the fully connected layers.

The models utilizing f_{iof} are extended versions of the models with f_{img} . These models incorporate additional features extracted using the OpenFace toolkit. The output of pre-trained models (ResNet18 and EfficientNet-B0) is concatenated with these features, and the combined feature vector is passed through a series of fully connected layers. This model explores how the combination of visual and facial feature data can enhance classification performance.

Additionally, experiments were conducted by freezing the convolutional layers of the ResNet and EfficientNet models. This technique aimed to retain the pre-trained knowledge from ImageNet while fine-tuning only the fully connected layers on the engagement classification task.

Similar to the approach with traditional machine learning models, experiments with deep learning models were conducted using different sets of OpenFace features: action units (AU), eye gaze, and head pose features. Separate configurations of deep learning models were trained and optimized. This experimentation enabled assessment of the impact of different feature sets on the models' performance, ensuring robust and well-generalized results.

Fig. 7 ROC AUC curves comparing the classification performance of SVM, random forest, decision tree, gradient boosting, and XGBoost models across four distinct datasets: S_1 (facial landmark features), S_2 (eye gaze and head pose features), S_3 (facial action unit features), and S_4 (composite feature set). Higher ROC AUC values indicate superior discriminative capabilities. SVM consistently exhibits the lowest performance across all datasets, while XGBoost and random forest consistently outperform other models

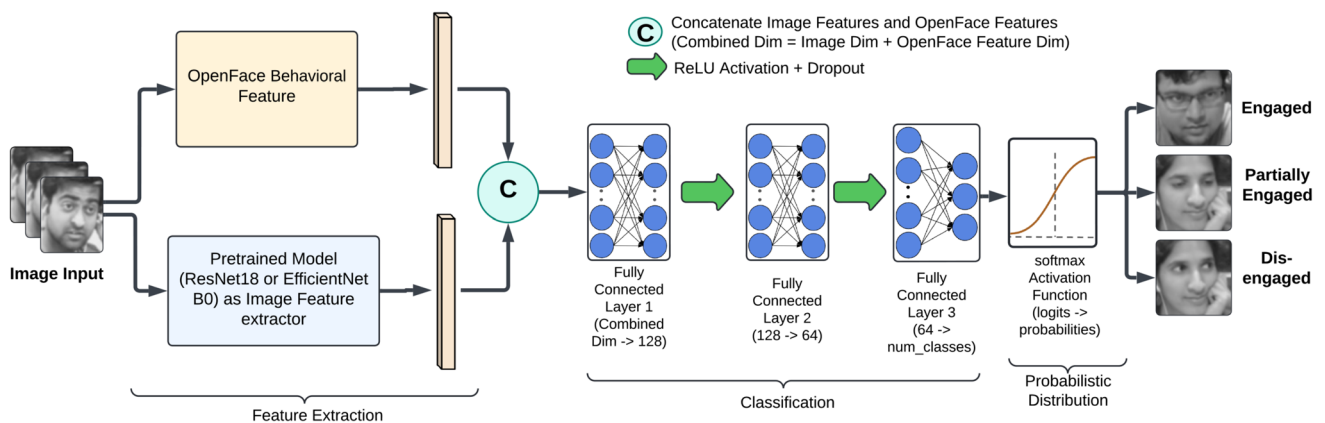
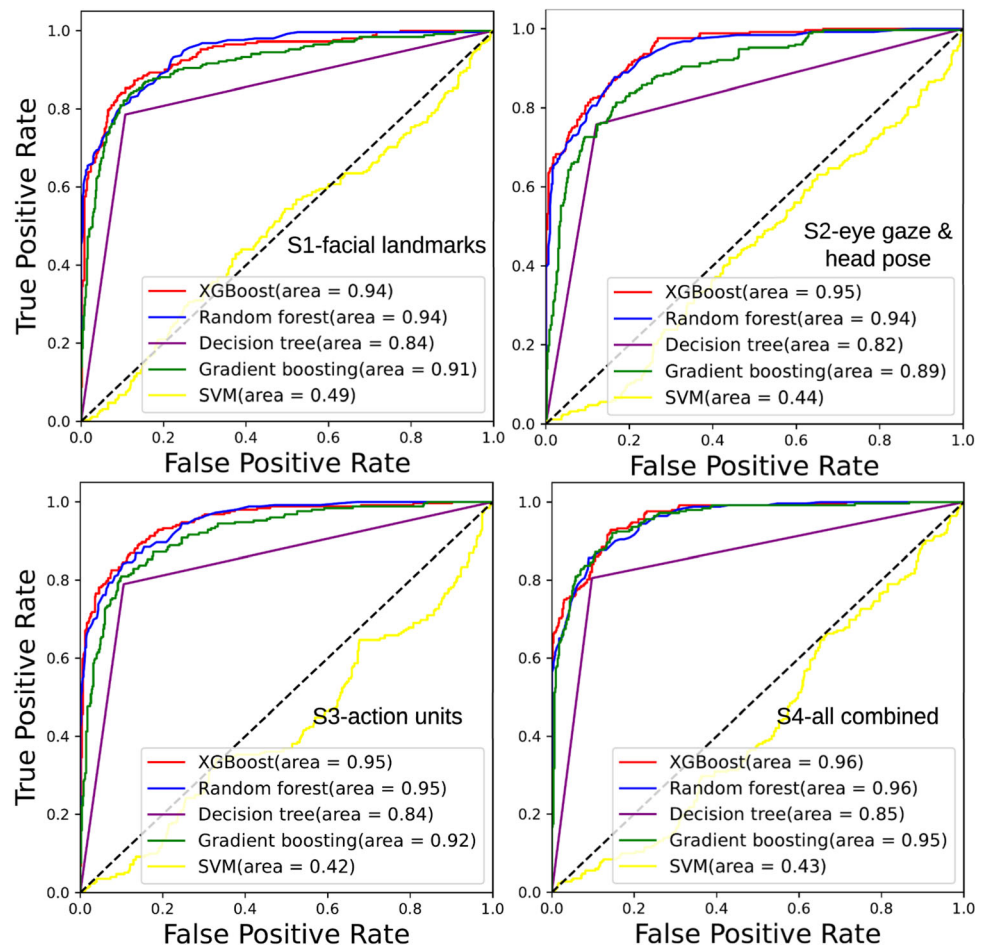


Fig. 8 Architecture of the f_{iof} model with input concatenation: the model combines image features from pre-trained ResNet/ EfficientNet with OpenFace features. These concatenated features are passed through fully connected layers with ReLU activation and dropout, leading to the final predictions for engagement levels

4.5.2 Dual-path feature

In the deep learning experiments, the potential of integrating visual data from images with facial landmark features was explored by implementing a dual-path feature model architecture, as illustrated in Fig. 9. This

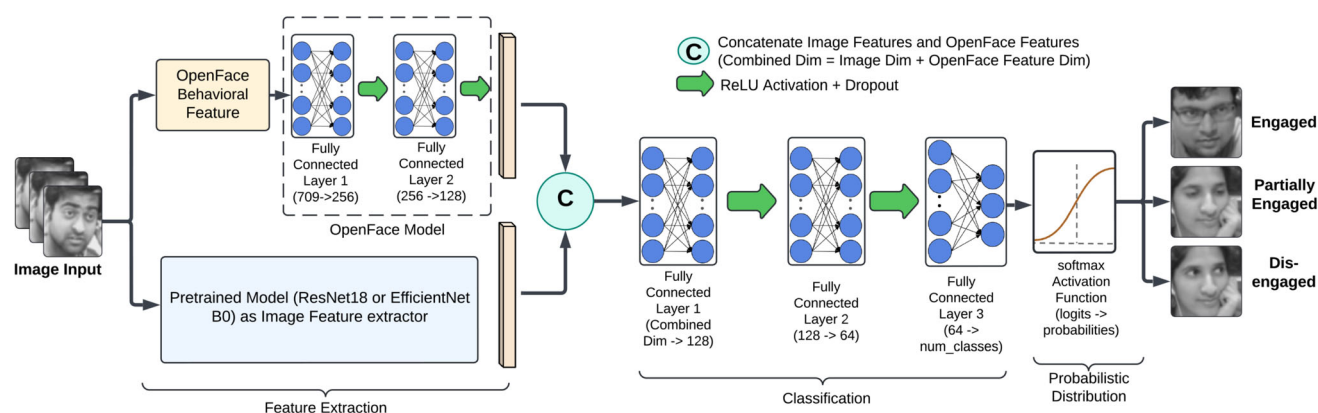


Fig. 9 Architecture of the dual-path model: the dual-path model integrates image features from either ResNet or EfficientNet with auxiliary features extracted by the OpenFace model. The OpenFace model processes these behavioral features through a multilayer perceptron (MLP) consisting of two fully connected layers, ReLU activations, and dropout for regularization, reducing the 709-dimensional input to a 128-dimensional feature vector. The extracted features from both the pre-trained image model and the OpenFace model are then combined and passed through additional fully connected layers, culminating in a final classification layer that predicts one of three classes

approach facilitated the harnessing of strengths from both types of input data images processed through convolutional neural networks (CNNs) and facial features extracted using the OpenFace toolkit.

This model utilizes pre-trained models (ResNet18 and EfficientNet-B0) on the ImageNet dataset as the base CNN for processing image data. The original fully connected layer of the model is replaced with an identity function, effectively transforming the network into a feature extractor. Simultaneously, facial features extracted by OpenFace were processed through a separate fully connected network.

The outputs from the feature extractor and the OpenFace feature network were concatenated, forming a combined feature vector that captures both visual and facial landmark information. This combined vector was then passed through a series of shared fully connected layers. These layers were responsible for learning the final classification boundaries across the three engagement classes. By sharing weights across the fully connected layers, the model could efficiently learn to integrate and interpret complementary information from both sources of input data.

Key benefits of these feature approaches (f_{iof}):

- By combining image and facial features before applying shared fully connected layers, the model can efficiently learn how these two data modalities complement each other, potentially leading to more accurate predictions.
- Sharing weights across fully connected layers can help reduce the risk of overfitting, especially in scenarios with limited data, as the model parameters are more tightly constrained.
- This approach leverages the strengths of both visual and facial feature data, providing a richer and more comprehensive input to the classification layers.

5 Discussion

5.1 Model interpretability and analysis

In this section, the performance and interpretability of deep learning models trained using a combination of image data and OpenFace features are examined, alongside models trained exclusively on image data. The analysis

Table 2 Evaluation metrics (accuracy, precision, recall, and F1-score) for different models trained with image data (f_{img}) and a combination of image data and OpenFace features (f_{iof}). The ResNet and EfficientNet models were evaluated on their ability to predict outcomes based on these input features

Model	Accuracy	Precision	Recall	F1-score
ResNet (f_{img})	0.301	0.388	0.301	0.297
ResNet (f_{iof})	0.365	0.404	0.365	0.298
EfficientNet (f_{img})	0.433	0.439	0.433	0.285
EfficientNet (f_{iof})	0.340	0.213	0.339	0.258

focuses on assessing the effectiveness of these models through a set of key evaluation metrics, including accuracy, precision, recall, and F1-score. To ensure a consistent and fair comparison, all models were trained and tested using identical data splits as those employed for the machine learning models discussed earlier.

The results presented in Table 2 indicate that models leveraging both image data and OpenFace features (f_{iof}) generally outperformed those relying solely on image data (f_{img}). Specifically, the ResNet model trained with combined features achieved an accuracy of 0.365, which is higher than the 0.301 accuracy observed for the ResNet model trained only on image data. Similarly, while the EfficientNet model with combined features exhibited a lower accuracy of 0.340 compared to its image-only counterpart (0.433), the model's overall performance across different metrics shows that the integration of OpenFace features adds interpretability and potentially captures more nuanced aspects of the data.

Further analysis was conducted by freezing the convolutional layers of the pre-trained models (ResNet and EfficientNet) to assess the impact of feature extraction from the pre-trained models on the performance metrics. As shown in Table 3, freezing the convolutional layers generally resulted in improved accuracy and precision across both models and feature sets. Notably, the ResNet model with combined features (f_{iof}) showed a significant improvement in accuracy, reaching 0.409, while the EfficientNet model with combined features achieved an accuracy of 0.421. This suggests that freezing the convolutional layers allows the models to focus on learning the specific high-level features relevant to the task at hand, leading to enhanced performance.

As mentioned in Sect. 4.5.2, the ResNet and EfficientNet models have been analyzed by utilizing a dual-path feature approach. In this methodology, image data and OpenFace features are processed independently through two distinct networks. The outputs of these networks are subsequently merged and passed through a series of shared fully connected layers, allowing the model to integrate the learned representations from both feature sets. The results of this dual-path approach are summarized in Table 4.

The ResNet model employing this dual-path configuration achieved an accuracy of 0.357, with corresponding precision, recall, and F1-score values of 0.459, 0.357, and 0.367, respectively. In comparison, the EfficientNet model, under the same dual-path feature setup, attained a higher accuracy of 0.421, though with slightly lower precision and F1-score values of 0.368 and 0.323, respectively.

Table 3 Evaluation metrics (accuracy, precision, recall, and F1-score) for different models after freezing () the convolutional layers of the pre-trained ResNet and EfficientNet models. The models were trained using image data alone (f_{img}) and a combination of image data and OpenFace features (f_{iof}) to assess the impact of feature extraction on performance

Model	Accuracy	Precision	Recall	F1-score
ResNet* (f_{iof})	0.409	0.362	0.408	0.366
ResNet* (f_{img})	0.332	0.436	0.332	0.331
EfficientNet* (f_{iof})	0.421	0.445	0.420	0.305
EfficientNet* (f_{img})	0.309	0.397	0.309	0.325

Table 4 Evaluation metrics (accuracy, precision, recall, and F1-score) for ResNet and EfficientNet models using the dual-path feature approach. This method processes image data and OpenFace features through separate networks, combining their outputs in a series of shared fully connected layers

Model	Accuracy	Precision	Recall	F1-score
ResNet	0.357	0.459	0.357	0.367
EfficientNet	0.421	0.368	0.421	0.323

To gain deeper insights into how the models arrive at their predictions, Grad-CAM, a visualization technique, has been employed that highlights the specific regions of the input image that have the most significant impact on the model's decision-making process. Figure 10 presents the Grad-CAM results for both the ResNet and EfficientNet architectures under two different feature configurations: with image-only inputs and with combined image and OpenFace features. For the EfficientNet model, the Grad-CAM visualizations illustrate the areas of focus across different blocks, specifically Blocks 6, 5, and 4. These blocks correspond to progressively earlier stages in the network, allowing us to observe how the model's attention evolves as the image is processed. This layered approach helps to uncover the sequential refinement of features, showing which parts of the face the model deems important at each stage. Similarly, for the ResNet model, the Grad-CAM results are provided for various layers, namely Layer 4 Block 2 Conv2, Layer 4 Block 1 Conv2, and Layer 3 Block 2 Conv2. These visualizations reveal how the model's focus shifts across different layers, from higher-level abstractions to more detailed feature representations. By analyzing these layers, it can be observed how ResNet progressively focuses on crucial facial features that contribute to its predictions. Across both models and feature configurations, the Grad-CAM outputs as shown in Fig. 10, highlight key facial regions, such as the eyes, mouth, and overall head orientation. These areas are vital for accurately interpreting facial expressions and are indicative of the model's understanding of the task. The visualizations not only demonstrate where the models are “looking,” but also provide insights into the hierarchical nature of feature extraction within these deep learning architectures, helping us to understand how different levels of abstraction are integrated to form the final decision.

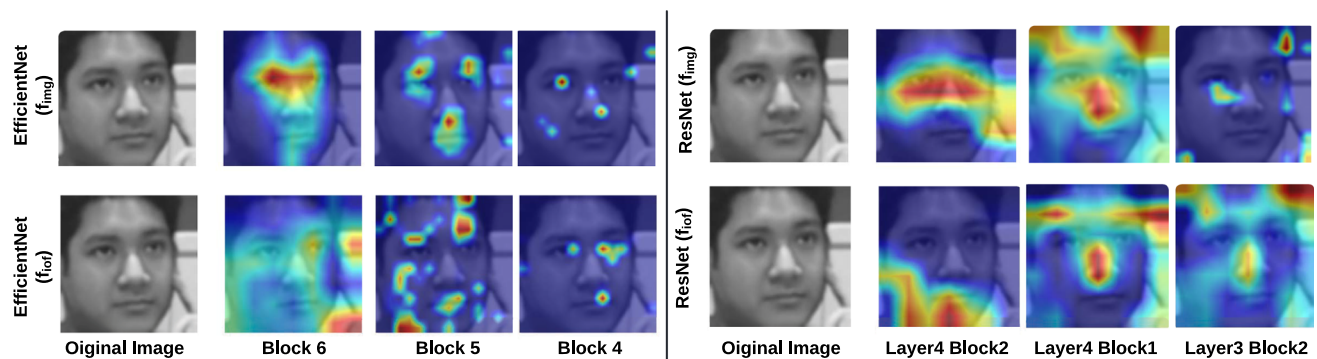


Fig. 10 Grad-CAM visualizations for EfficientNet and ResNet architectures under two feature configurations: f_{img} (image-only) and f_{iof} (image and open-face features). For EfficientNet, the visualizations are shown for Block 6, Block 5, and Block 4, both for f_{img} and f_{iof} . Similarly, for ResNet, the visualizations are provided for Layer 4 Block 2 Conv2, Layer 4 Block 1 Conv2, and Layer 3 Block 2 Conv2 for both f_{img} and f_{iof} . These Grad-CAM outputs highlight the specific regions of the images that the models focus on at various layers, offering insights into the decision-making process at different levels of abstraction within the networks

5.2 Ablation study

The evaluation of the models revealed significant insights into the role of incorporating auxiliary features, such as action units (AU), eye gaze, and head pose, in enhancing model performance. To isolate the impact of different feature sets, an ablation study was conducted systematically by removing the OpenFace features and observing the resulting changes in performance. Table 5 summarizes the performance metrics of the ResNet and EfficientNet models when combining image data with all OpenFace features (f_{iof}), as well as with specific subsets of OpenFace features, such as action units (f_{AU}) and eye gaze and head pose (f_{EH}). The results clearly demonstrate that integrating action unit features significantly improves performance across all metrics (accuracy, precision, recall, and F1-score) compared to models without these features. For instance, both the ResNet and EfficientNet models incorporating action units (f_{AU}) achieved superior results compared to their counterparts relying solely on image data or other OpenFace features. This ablation study highlights the effectiveness of a multimodal approach, showing that combining image data with auxiliary features substantially enhances the model's ability to generalize across diverse scenarios, leading to more robust and accurate predictions.

5.3 Hyperparameter analysis

Hyperparameter optimization is a critical step in enhancing the performance of machine learning models. In this study, Optuna, a state-of-the-art hyperparameter optimization framework, was leveraged to tune the models for optimal performance across various tasks.

We performed hyperparameter optimization across several model configurations, including both ResNet and EfficientNet architectures, under different feature settings such as f_{img} , f_{iof} , and freeze variations. This optimization was conducted individually for each model and feature combination, leading to a set of optimal hyperparameters tailored to each specific scenario. The comprehensive list of the best hyperparameters obtained for the various model settings is detailed in Table 6.

To provide insight into the importance of different hyperparameters across models, the parameter importance was visualized using horizontal bar plots in Fig. 11. Each model's parameter importance is presented with a distinct color, facilitating a comparative analysis of how different hyperparameters contribute to the models' performance. This visual representation allows us to observe which parameters are consistently influential across various configurations and it helps in understanding the sensitivity of models to different hyperparameters, guiding future model refinement and tuning efforts.

6 Comparative analysis of AU mapping in static and dynamic data

From a static image, it is not always possible to fully understand the mapping of AU with engagement labels. In this work, we have mainly performed our engagement analysis on the WACV dataset, which consists only of image data, not video clips. Therefore, as an extension, we have performed the engagement analysis on the DAiSEE [31] dataset. The DAiSEE dataset consists of 12583 video clips each 10sec long. Each of these video clips is annotated [32] with

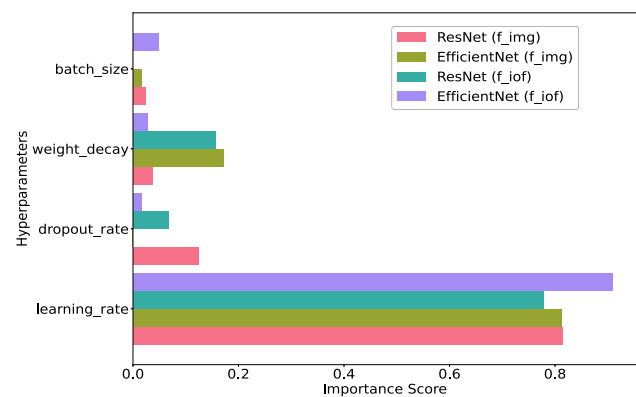
Table 5 Performance metrics of ResNet and EfficientNet models integrating image data with all OpenFace features (f_{iof}), as well as specific subsets like action units (f_{AU}) and eye gaze and head pose (f_{EH})

Model	Accuracy	Precision	Recall	F1 Score
ResNet (f_{iof})	0.409	0.362	0.408	0.366
ResNet (f_{AU})	0.437	0.468	0.437	0.374
ResNet (f_{EH})	0.234	0.353	0.234	0.242
EfficientNet (f_{iof})	0.421	0.445	0.420	0.305
EfficientNet (f_{AU})	0.472	0.229	0.472	0.309
EfficientNet (f_{EH})	0.433	0.464	0.433	0.350

Table 6 Best hyperparameter values for various models, including ResNet and EfficientNet variations, across different configurations. The table presents the optimal values for batch size, learning rate, dropout rate, and weight decay for each model.

Model	Best Hyperparameters			
	Batch size	Learning rate	Dropout rate	Weight decay
ResNet f_{AU}	64	$2.12e^{-4}$	0.2501	$2.45e^{-5}$
EfficientNet f_{AU}	64	$1.42e^{-4}$	0.4411	$1.50e^{-4}$
ResNet f_{EH}	64	$4.51e^{-5}$	0.4283	$5.58e^{-5}$
EfficientNet f_{EH}	32	$1.32e^{-4}$	0.3716	$3.90e^{-6}$
ResNet f_{img}	64	$2.54e^{-4}$	0.4982	$4.40e^{-5}$
EfficientNet f_{img}	64	$1.58e^{-4}$	0.2175	$3.07e^{-4}$
ResNet f_{iof}	32	$3.32e^{-4}$	0.3482	$1.24e^{-5}$
EfficientNet f_{iof}	32	$9.39e^{-5}$	0.4140	$4.95e^{-6}$
ResNet f_{img} (freeze)	16	$1.03e^{-5}$	0.4986	$2.15e^{-4}$
EfficientNet f_{img} (freeze)	16	$5.88e^{-5}$	0.4600	$1.11e^{-6}$
ResNet f_{iof} (freeze)	16	$1.98e^{-5}$	0.2477	$1.18e^{-4}$
EfficientNet f_{iof} (freeze)	32	$1.81e^{-5}$	0.4053	$6.52e^{-4}$

Fig. 11 Importance of different hyperparameters across various models. Each bar represents the importance score of a hyperparameter, with different colors denoting distinct models.



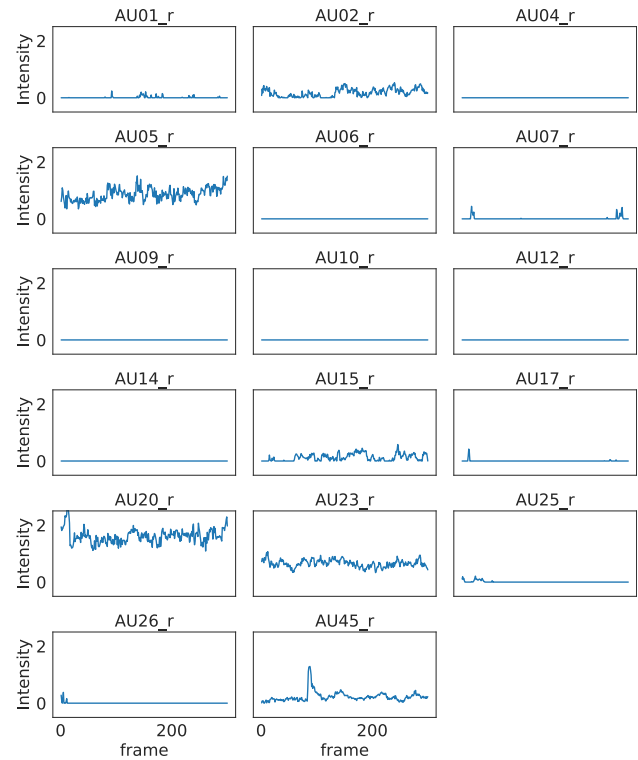
different labels, boredom, engagement, confusion, and frustration. Concerning engagement analysis, we are interested only in the engagement label. The engagement label is marked from 1 to 3, where 1 represents low engagement and 3 represents very high engagement. We generated facial features (facial landmark, eye gaze, head pose, and AU features) using OpenFace for each 10-sec-long video. From an input video, a maximum of 300 frame sequences are generated, and for each such frame, 712 facial features are generated. Figure 12 shows different intensity levels of 17 action units for each video frame. From the figure, we can see that several AUs (e.g., AU14, AU04) have zero intensity throughout the frames. AUs like AU17 and AU26 exhibit a peak value within the starting 50 frames.

Understanding the mapping between action units (AUs) and engagement labels presents unique challenges in static images versus dynamic videos. While static image datasets such as the WACV dataset offer stable facial representations, dynamic datasets like DAiSEE capture the temporal evolution of facial expressions. These differences have notable implications for AU engagement mapping.

In the WACV dataset, each image represents a discrete instance with a stable facial expression. Consequently, AU intensities are extracted directly from the image, and a single engagement label is assigned to the instance. In this context, the conditional probability $P_{ij} = P(AU_j|y_i)$ is computed as the fraction of images labeled y_i in which AU_j is activated. Additionally, the relative activation ratio for AU_j in engagement class y_i is computed as:

$$R_{ij} = \frac{P(AU_j|y_i)}{P(AU_j)} \quad (7)$$

Fig. 12 Intensity values (in the range of 0 to 1) of the different AUs throughout the 300 frames of the input video



where $P(AU_j)$ is the overall activation probability of AU j across all classes. This coefficient identifies AUs that are disproportionately activated in specific engagement states. The log-ratio statistical discriminative coefficient (SDC) is then calculated as follows:

$$S_j = \sum_i P(y_i) \log \left(\frac{P(AU_j|y_i)}{P(AU_j)} \right) \quad (8)$$

where S_j quantifies the overall discriminative power of AU j across all engagement labels. Higher SDC values indicate stronger class-specific behavior.

In contrast, the DAiSEE dataset comprises video sequences, where each video is annotated with a single engagement label. For analysis purposes, AU features are extracted frame-by-frame, and the engagement label is assigned uniformly across all frames. This transformation effectively mimics the static image structure while preserving the temporal nature of the data. Consequently, each video provides a sequence of AU intensity values, capturing dynamic variations in facial behavior.

The inherent temporal variability in videos introduces challenges absent in static datasets. Transient facial expressions, micro-expressions, and brief emotional shifts can create fluctuations in AU intensities. As a result, the per-frame conditional probabilities $P_{ij} = P(AU_j|y_i)$ may exhibit greater variability, particularly for AUs that are sensitive to rapid facial changes. This variability directly affects the relative activation ratios R_{ij} and SDC values, which may fluctuate more extensively in video data compared to static images.

Figure 13 illustrates the overall activation rates for 17 action units (AUs) in the DAiSEE dataset, providing insights into the frequency of AU activations across video frames. The heatmap reveals distinct patterns, highlighting AUs that consistently appear in facial expressions across engagement states. Notably, AU05, AU20, and AU14 exhibit the highest overall activation rates at 0.60, 0.64, and 0.54, respectively. These AUs are prominent across multiple engagement states, reinforcing their significance in general facial behavior. Conversely, AU02, AU10, and AU26 show lower activation rates (all below 0.15), indicating their infrequent presence in the dataset.

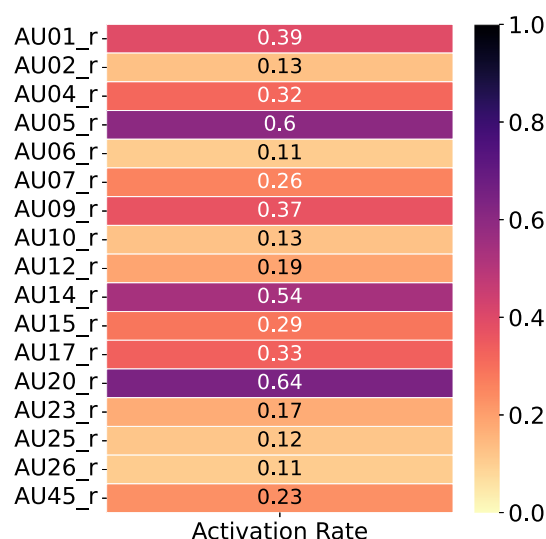


Fig. 13 Overall AU activation rates in DAiSEE dataset. This heatmap illustrates the proportion of frames in which each action unit (AU) is activated across the video dataset. A higher value indicates that the corresponding AU is active more frequently. Notably, AUs such as **AU05** (0.60), **AU20** (0.64), and **AU14** (0.54) exhibit the highest overall activation rates, suggesting their consistent involvement in facial expressions across multiple engagement states

This aligns with the dynamic nature of video data, where subtle or transient facial expressions may not sustain high AU intensity values throughout a 10-second video sequence.

7 Conclusion and future work

In this study, both classical machine learning models and advanced deep learning architectures were employed to analyze engagement using facial features extracted from the WACV dataset. Initially, using tools such as OpenFace and MediaPipe, facial features were extracted and various machine learning models—including SVM, random forest, decision tree, gradient boosting, and XGBoost—were trained. The evaluation demonstrated that XGBoost and gradient boosting consistently outperformed other models across metrics such as accuracy, precision, recall, and F-measure.

Further extending the analysis, deep learning models—specifically ResNet and EfficientNet—were introduced into the framework. These models, when trained on a combination of image data and OpenFace features, exhibited superior performance in predicting engagement labels. The models demonstrated an enhanced ability to capture intricate facial details and variations, leading to more accurate classification of engagement levels.

Moreover, specific action units (AUs) were mapped to engagement labels within the WACV dataset, identifying predictive AU combinations using the top three statistical discriminant coefficients (SDC) values. For example, AU14, AU23, and AU20 were found to be indicative of high engagement, AU05, AU15, and AU20 for partial engagement, and AU05, AU45, and AU20 for disengagement. These mappings were then utilized to enhance the performance of both machine learning and deep learning models, allowing for more precise prediction and recognition of engagement labels from image and video data.

Looking ahead, this work aims to be extended by applying these findings to real-time video data, advancing toward the automation of engagement recognition processes. Specifically, temporal deep learning techniques, such as LSTM networks and TCNs, are planned to be integrated to better capture the dynamic nature of engagement as it evolves over time. Additionally, valence-arousal modeling [33] is intended to be included to enhance the detection of facial emotions, providing a more nuanced understanding of emotional states that contribute to engagement levels. This will facilitate the development of real-time systems capable of detecting

and interpreting engagement levels in continuous video streams, with potential applications [34] in areas such as online education, remote work, and human–computer interaction.

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Author contributions R. Das conducted the experiments and wrote the manuscript text. S. Dev supervised and also wrote the manuscript text. All authors reviewed the manuscript.

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Data availability The data that support the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

Ethical approval Not applicable.

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